SOA IDEATHON 2026 PROBLEM STATEMENTS [SOFTWARE & HARDWARE EDITIONS]				
Problem Statement ID	**Title**	**Description**	**Category**	**Theme**
SOAIDEATHON-S1	Human-in-the-Loop Agentic AI for Autonomous Institutional Service Delivery	Develop a secure agentic-AI platform that can understand service requests, plan multi-step actions, retrieve verified institutional information, route approvals and execute permitted workflows such as certificate requests, maintenance tickets, laboratory bookings and grievance escalation. The system must maintain an auditable action trail, ask for human approval before consequential actions, support multilingual interaction and detect uncertainty or policy conflicts instead of fabricating answers.	Software	Smart Automation
SOAIDEATHON-H1	Digital Twin and Edge-AI Predictive Maintenance for Campus Laboratories and Utilities	Create a low-cost sensor network and digital twin for monitoring laboratory equipment, elevators, water pumps, HVAC systems and electrical panels. Edge-AI models should identify abnormal vibration, temperature, power consumption or usage patterns, estimate remaining useful life and generate maintenance recommendations. The solution must continue functioning during internet outages, prioritize critical assets and provide explainable alerts to technicians.	Hardware	Smart Automation
SOAIDEATHON-S2	Multilingual Regulatory Document Intelligence and Compliance Workflow Engine	Develop an AI system that ingests circulars, accreditation requirements, procurement rules and institutional policies; extracts obligations, deadlines, responsible units and evidence requirements; and converts them into traceable tasks. It should compare new documents with earlier versions, flag conflicting clauses, generate compliance dashboards and cite the exact source passage for every recommendation. Sensitive documents must be processed with role-based access and data-retention controls.	Software	Smart Automation
SOAIDEATHON-S3	Self-Healing AI Operations Controller for Campus Digital and Physical Infrastructure	Develop a human-governed AIOps platform that correlates network logs, application telemetry, facility alarms and IoT sensor streams to identify the likely root cause of service degradation and recommend safe recovery playbooks. The system should rank actions by risk, simulate or sandbox changes before execution, maintain a complete audit trail and require explicit approval for high-impact actions. It must continue to provide useful diagnostics when some sensors or services are unavailable.	Software	Smart Automation

SOAIDEATHON-SH1	**Student Innovation – Smart Automation: Open Challenge**	**Identify and solve a high-impact real-world problem related to intelligent automation, AI-assisted operations, resource optimization or decision support that is not already addressed by the four institution-defined problem statements listed under the Smart Automation theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Smart Automation**
SOAIDEATHON-H2	Multimodal Athlete Talent Assessment and Injury-Risk Prevention Kit	Design a portable kit using smartphone video, wearable inertial sensors and force or pressure measurements to assess movement quality, endurance and sport-specific skills. AI models should generate age- and sport-appropriate benchmarks, identify asymmetry or unsafe technique and recommend progressive training. The system must work in low-connectivity environments and avoid making medical diagnoses.	Hardware	Fitness & Sports
SOAIDEATHON-S4	Inclusive Adaptive Coaching Platform for Athletes with Disabilities	Develop an accessible coaching platform that creates personalized training plans for athletes with visual, hearing, mobility or cognitive disabilities. It should support voice, sign-language video, haptic prompts, simplified interfaces and caregiver or coach dashboards. Progress recommendations must adapt to available equipment, fatigue and accessibility needs while allowing users to control what health and performance data are shared.	Software	Fitness & Sports
SOAIDEATHON-S5	Community Sports Facility Utilization, Safety and Energy Optimization System	Create a platform that forecasts demand for courts, fields and gyms, manages fair time-slot allocation, detects unsafe overcrowding and recommends energy-efficient operating schedules. The system should integrate booking data, occupancy sensors and weather information while ensuring transparent access rules for students, women, persons with disabilities and community users.	Software	Fitness & Sports

SOAIDEATHON-H3	Edge-AI Fair Officiating and Performance Analytics Kit for Grassroots Sports	Design a portable multi-camera and edge-AI kit that assists referees and coaches in grassroots sports by detecting rule-relevant events, player positions and timing while preserving the referee as the final decision-maker. The system should provide replayable evidence, quantify confidence, work with modest cameras and intermittent internet, and include bias testing across venues, lighting conditions and participant groups. It should also generate post-match performance indicators without continuous biometric surveillance.	Hardware	Fitness & Sports
SOAIDEATHON-SH2	**Student Innovation – Fitness & Sports: Open Challenge**	**Identify and solve a high-impact real-world problem related to fitness participation, athlete safety, sports access, coaching, officiating or community sports that is not already addressed by the four institution-defined problem statements listed under the Fitness & Sports theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Fitness & Sports**
SOAIDEATHON-S6	Edge-AI Earth Observation Platform for Hyperlocal Flood, Crop and Heat-Stress Alerts	Develop a platform that fuses satellite imagery, weather feeds and local IoT observations to generate ward- or village-level alerts for flood extent, crop stress and urban heat. Models should quantify uncertainty, explain the evidence behind each alert and provide lightweight outputs suitable for low-bandwidth mobile devices. The system must allow local authorities to validate or correct alerts through field observations.	Software	Space Technology
SOAIDEATHON-H4	GNSS-Denied Visual-Inertial Navigation Module for Emergency Drones and Ground Robots	Design a compact navigation module that enables drones or ground robots to localize and return safely when GNSS signals are unavailable, jammed or unreliable. The system should fuse camera, inertial and optional LiDAR data; recognize degraded conditions; and switch to a fail-safe mode. Performance must be demonstrated in indoor, urban-canyon and smoke-obscured test environments.	Hardware	Space Technology

SOAIDEATHON-H5	Low-Cost AI-Assisted Ground Station for CubeSat Telemetry Quality and Anomaly Detection	Develop a modular ground-station kit that receives educational satellite or simulated CubeSat telemetry, automatically assesses signal quality, identifies anomalous subsystem behavior and visualizes pass predictions. The platform should include a digital twin for generating fault scenarios, support student experiments and publish interoperable telemetry APIs without exposing protected command functions.	Hardware	Space Technology
SOAIDEATHON-H6	Onboard Edge-AI Data Prioritization and Adaptive Downlink for Small Satellites	Develop an onboard-computing prototype that classifies and prioritizes satellite imagery or sensor packets before downlink when bandwidth, power or ground-station contact time is limited. The solution should detect low-value or corrupted data, retain scientifically important observations, adapt compression and transmission priority to mission rules, and provide an explainable record of what was kept, deferred or discarded. A hardware-in-the-loop simulator may be used where satellite hardware is unavailable.	Hardware	Space Technology
SOAIDEATHON-SH3	**Student Innovation – Space Technology: Open Challenge**	**Identify and solve a high-impact real-world problem related to satellite systems, earth observation, space communication, navigation, payloads or exploration support that is not already addressed by the four institution-defined problem statements listed under the Space Technology theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Space Technology**
SOAIDEATHON-S7	Provenance-Aware AI Archive for India's Oral Histories and Endangered Languages	Create a multilingual platform to record, transcribe, translate and index oral histories, folk knowledge and endangered-language narratives. Every AI-generated transcript or translation must retain links to the original audio, speaker consent, community-approved access restrictions and version history. The system should help researchers discover themes without allowing synthetic content to be confused with authentic testimony.	Software	Heritage & Culture

SOAIDEATHON-H7	AR Digital Twin and Non-Contact Condition Monitoring for Local Heritage Structures	Develop a mobile and sensor-assisted system that creates an AR-ready 3D digital twin of a monument or historic building and detects cracks, moisture, surface loss or unauthorized changes over time. The system should compare periodic imagery, prioritize inspection zones and present restoration history to visitors without physically altering the site.	Hardware	Heritage & Culture
SOAIDEATHON-S8	Authenticity and Fair-Trade Digital Passport for Traditional Crafts	Create a traceability platform through which artisans can issue tamper-evident digital product passports containing origin, materials, techniques, maker attribution and care instructions. Buyers should be able to verify authenticity through a low-cost tag, while cooperatives can track fair compensation and detect counterfeit listings. The solution must remain usable for artisans with limited digital literacy.	Software	Heritage & Culture
SOAIDEATHON-S9	AI-Assisted Restoration and Scholarly Reconstruction of Damaged Manuscripts and Inscriptions	Create a scholar-in-the-loop platform that enhances faded manuscript pages, palm-leaf records, inscriptions or archival documents; proposes alternative character or word reconstructions; and links each suggestion to visual evidence and reference corpora. The tool must preserve the original image, mark uncertain restorations, allow experts to accept or reject suggestions and maintain provenance for every edit. It should support at least one Indian script and one local collection.	Software	Heritage & Culture
SOAIDEATHON-SH4	**Student Innovation – Heritage & Culture: Open Challenge**	**Identify and solve a high-impact real-world problem related to preservation, documentation, interpretation, language, crafts, archives or cultural participation that is not already addressed by the four institution-defined problem statements listed under the Heritage & Culture theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Heritage & Culture**

SOAIDEATHON-S10	Federated Community Health Early-Warning System for Emerging Disease Clusters	Develop a privacy-preserving system that learns from de-identified symptom trends, campus or clinic visits, water-quality indicators and local environmental data without centralizing personal health records. It should identify unusual disease clusters, estimate confidence, explain contributing signals and notify authorized public-health users while minimizing false alarms and protecting small groups from re-identification.	Software	MedTech/BioTech/ HealthTech
SOAIDEATHON-H8	Non-Invasive Multimodal Screening Kiosk for Anaemia and Respiratory Risk	Design a low-cost screening kiosk using approved non-invasive sensors, smartphone imaging and questionnaires to identify people who may need confirmatory testing for anaemia or respiratory risk. The system should provide uncertainty-aware referral guidance, work offline and be calibrated across diverse skin tones and environmental conditions. It must clearly state that it is a screening aid, not a diagnostic device.	Hardware	MedTech/BioTech/ HealthTech
SOAIDEATHON-S11	Explainable Antimicrobial Stewardship and Prescription Safety Assistant	Develop a clinical decision-support prototype that checks prescriptions against diagnosis, allergies, renal considerations, duplication, local antimicrobial guidelines and resistance patterns. It should highlight the source rule for every warning, allow clinician override with reason capture and generate anonymized stewardship analytics. The platform must not autonomously prescribe or replace clinical judgment.	Software	MedTech/BioTech/ HealthTech
SOAIDEATHON-H9	Privacy-Preserving Wearable Gait, Balance and Fall-Risk Assessment for Elderly and Neuro-Rehabilitation	Design a low-cost wearable and mobile system using inertial sensing and optional pressure data to quantify gait, balance and rehabilitation exercises in home or community settings. The system should detect meaningful changes, explain which movement features drive the risk or progress score, operate offline, avoid raw-video dependence and allow clinicians or therapists to configure thresholds. It must be framed as a monitoring and referral aid, not an autonomous diagnosis.	Hardware	MedTech/BioTech/ HealthTech

SOAIDEATHON-SH5	**Student Innovation – MedTech/BioTech/ HealthTech: Open Challenge**	**Identify and solve a high-impact real-world problem related to health prevention, screening, care delivery, assistive technology, biotechnology or public health that is not already addressed by the four institution-defined problem statements listed under the MedTech/BioTech/ HealthTech theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**MedTech/BioTech/ HealthTech**
SOAIDEATHON-S12	Climate-Resilient Farm Digital Twin with Explainable Crop, Water and Input Advisory	Create a farm digital twin that combines soil sensors, local weather, satellite observations, crop stage and farmer records to recommend irrigation, crop choice, nutrient application and risk-mitigation actions. Recommendations must include confidence, expected cost-benefit and the factors that influenced them. The platform should support local languages, offline synchronization and farmer feedback.	Software	Agriculture, FoodTech & Rural Development
SOAIDEATHON-H10	Vision-and-Sensor Post-Harvest Quality Grading and Spoilage Prediction Unit	Design an affordable unit that grades fruits, vegetables or grains using visible or multispectral imaging, temperature, humidity and gas sensing. The system should estimate remaining shelf life, recommend storage actions and provide a traceable quality report for farmers, aggregators and buyers. It must be calibrated for one selected commodity and operate in rural pack-house conditions.	Hardware	Agriculture, FoodTech & Rural Development
SOAIDEATHON-H11	Cooperative Autonomous Micro-Machine for Precision Weeding and Shared Rural Use	Develop a small, repairable autonomous machine for precision weeding or spot treatment on small farms. It should navigate uneven fields, identify crop rows, avoid people and animals, and log completed work. A digital booking and maintenance layer should enable shared ownership through farmer groups rather than requiring individual purchase.	Hardware	Agriculture, FoodTech & Rural Development

SOAIDEATHON-H12	Autonomous Pest and Disease Sentinel Network with Precision Bio-Intervention Guidance	Develop a distributed field sentinel using low-cost cameras, pheromone or environmental sensors and edge AI to detect early pest or disease signatures before visible crop loss becomes widespread. The system should map hotspots, estimate confidence, recommend integrated pest-management actions and, where appropriate, trigger only localized non-chemical or biological intervention through an approved actuator. It must minimize unnecessary pesticide use and support local-language farmer alerts.	Hardware	Agriculture, FoodTech & Rural Development
SOAIDEATHON-SH6	**Student Innovation – Agriculture, FoodTech & Rural Development: Open Challenge**	**Identify and solve a high-impact real-world problem related to agriculture, food processing, rural livelihoods, farm mechanization, agri supply chains or resource efficiency that is not already addressed by the four institution-defined problem statements listed under the Agriculture, FoodTech & Rural Development theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Agriculture, FoodTech & Rural Development**
SOAIDEATHON-H13	Cooperative Near-Miss Prevention System for Two-Wheelers, Buses and School Zones	Design a vehicle-to-everything safety system that uses low-cost roadside and onboard sensing to predict near-miss situations involving two-wheelers, pedestrians and buses. It should issue timely warnings without distracting drivers, continue operating with intermittent connectivity and protect participant identities. Evaluation should focus on conflict reduction rather than only collision detection.	Hardware	Smart Vehicles
SOAIDEATHON-S13	Explainable EV Battery Health, Safety and Second-Life Digital Passport	Develop a platform that estimates battery state of health and safety risk from charging history, temperature, usage and diagnostic data. It should explain degradation factors, recommend safe charging practices and create a verifiable second-life passport for reuse or recycling. The system must distinguish estimation from certified testing and support multiple battery formats.	Software	Smart Vehicles

SOAIDEATHON-H14	Accessible Low-Speed Autonomous Shuttle for Campuses and Controlled Communities	Develop a low-speed electric shuttle prototype or digital-twin demonstrator that supports autonomous navigation on mapped routes, wheelchair boarding assistance, obstacle detection and remote human supervision. The system should recognize uncertainty, stop safely when conditions exceed capability and provide accessible passenger information through audio, visual and haptic cues.	Hardware	Smart Vehicles
SOAIDEATHON-H15	Privacy-Preserving In-Cabin Radar and Edge-AI Safety Assistant for Public and School Vehicles	Design an in-cabin safety system that uses non-contact radar, inertial sensing and privacy-preserving edge analytics to detect driver drowsiness, unsafe distraction, unattended occupants and abnormal cabin conditions without storing identifiable video. The system should issue graded warnings, minimize nuisance alerts, operate during poor lighting and intermittent connectivity, and retain only event evidence required for safety review. It should include safeguards against using the system for unauthorized surveillance.	Hardware	Smart Vehicles
SOAIDEATHON-SH7	**Student Innovation – Smart Vehicles: Open Challenge**	**Identify and solve a high-impact real-world problem related to vehicle safety, intelligent mobility, EV systems, in-vehicle technology or connected/autonomous vehicles that is not already addressed by the four institution-defined problem statements listed under the Smart Vehicles theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Smart Vehicles**
SOAIDEATHON-S14	AI-Optimized Rural Last-Mile Network for Farm Produce, Medicines and Essential Goods	Develop a logistics platform that dynamically combines available vehicles, community pickup points and delivery priorities to serve remote areas. It should optimize routes for perishability, urgency, road condition, vehicle capacity and cost; support offline updates; and provide transparent allocation so small producers and vulnerable communities are not consistently deprioritized.	Software	Transportation & Logistics
SOAIDEATHON-S15	Digital Twin for Public-Transport Disruptions, Crowd Flow and Emergency Rerouting	Create a digital twin of a local bus, metro or campus transport network that predicts crowding and service disruption from live or simulated vehicle, ticketing and event data. It should test alternate schedules, communicate accessible rerouting options and quantify passenger delay, safety and energy impacts before operators act.	Software	Transportation & Logistics

SOAIDEATHON-S16	Carbon-Aware Reverse Logistics and Reuse Marketplace for Institutional Assets	Develop a platform that tracks surplus equipment, packaging, electronics and furniture across departments; recommends reuse, repair, redistribution or certified recycling; and optimizes collection routes. It should calculate avoided purchase, waste and carbon impacts while preserving data-wiping and asset-approval requirements.	Software	Transportation & Logistics
SOAIDEATHON-S17	AI-Based Multimodal Freight Consolidation and Cold-Chain Risk Prediction for MSME and Agri Logistics	Develop a logistics decision platform that consolidates small shipments across road, rail or local transport options while predicting spoilage and delay risk for temperature-sensitive goods. It should recommend shipment grouping, departure time, route and transfer points using capacity, temperature history, road/rail reliability, cost and service-level constraints. The platform must protect commercially sensitive data and explain why a particular consolidation plan was selected.	Software	Transportation & Logistics
SOAIDEATHON-SH8	**Student Innovation – Transportation & Logistics: Open Challenge**	**Identify and solve a high-impact real-world problem related to public transport, freight, last-mile delivery, traffic operations, logistics or mobility infrastructure that is not already addressed by the four institution-defined problem statements listed under the Transportation & Logistics theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Transportation & Logistics**
SOAIDEATHON-H16	Resilient Swarm of Drones for Search, Mapping and Emergency Communication	Develop a small drone-swarm system that collaboratively maps an affected area, searches for people or hazards and forms a temporary communication relay when normal networks fail. The swarm should tolerate the loss of a drone, avoid no-fly areas, share confidence-tagged observations and permit human command override at all times.	Hardware	Robotics and Drones
SOAIDEATHON-H17	Soft Robotic Inspection and Cleaning System for Sewers and Hazardous Confined Spaces	Design a modular soft or compliant robot that can move through irregular pipes or confined spaces, identify blockages, gas risks or structural damage and perform limited cleaning or sample collection. The system should protect operators from direct exposure, recover safely after communication loss and generate an inspection map.	Hardware	Robotics and Drones

SOAIDEATHON-H18	Autonomous Laboratory Inventory and Safety-Audit Robot	Develop a mobile robot that scans equipment and consumable tags, verifies storage conditions, identifies blocked exits or misplaced safety items and updates a laboratory inventory. The robot should navigate around people, respect restricted zones and escalate uncertain findings for human verification rather than taking unsafe actions.	Hardware	Robotics and Drones
SOAIDEATHON-H19	Human-Safe Collaborative Robot for Disaster Debris Sorting and Victim-Safe Access	Design a teleoperated or semi-autonomous robotic platform that can enter unstable debris zones, classify light debris, detect signs of human presence and move selected obstructions without increasing risk to trapped persons or responders. The robot should combine force sensing, vision and acoustic or thermal cues, stop safely under uncertainty and provide the operator with a clear 3D or map-based view of actions. Autonomous movement of heavy debris should be prohibited unless validated in a controlled environment.	Hardware	Robotics and Drones
SOAIDEATHON-SH9	**Student Innovation – Robotics and Drones: Open Challenge**	**Identify and solve a high-impact real-world problem related to robots, drones, autonomous inspection, rescue, service robotics or human-robot collaboration that is not already addressed by the four institution-defined problem statements listed under the Robotics and Drones theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Robotics and Drones**
SOAIDEATHON-H20	AI-Assisted Material Recovery Station for Source-Level Waste Segregation	Design a compact material-recovery station that identifies common waste categories, detects contamination and guides users to the correct bin through visual, audio or tactile feedback. It should quantify segregation quality, learn from locally generated waste and support human correction. The mechanical design must be safe, easy to clean and maintainable by institutional staff.	Hardware	Clean & Green Technology

SOAIDEATHON-S18	Hyperlocal Urban Heat and Air-Quality Digital Twin for Cooling Intervention Planning	Create a platform that combines low-cost sensors, satellite data, building form, vegetation and mobility patterns to map heat and air-quality exposure at street or campus scale. It should simulate interventions such as shade, cool roofs, tree placement and traffic changes, prioritizing benefits for vulnerable users and quantifying uncertainty.	Software	Clean & Green Technology
SOAIDEATHON-H21	Closed-Loop Grey-Water, Rainwater and Non-Revenue Water Management System	Develop a sensor-assisted system that monitors water inflow, consumption, leakage, grey-water quality, treatment and reuse potential. It should detect abnormal losses, recommend safe reuse applications and provide a digital water balance with alerts for maintenance. The design must include safeguards that prevent unsuitable water from being routed for human contact.	Hardware	Clean & Green Technology
SOAIDEATHON-H22	Autonomous Microplastic and Urban Water-Pollutant Sensing Network with Source Attribution	Develop a low-cost network of fixed or mobile sensing nodes that screens drains, ponds or campus water bodies for microplastic proxies and selected chemical pollution indicators, combines measurements with flow and rainfall data, and identifies likely pollution hotspots or source zones. The system should quantify uncertainty, support confirmatory laboratory sampling and visualize trends without claiming laboratory-grade identification when the sensor cannot provide it.	Hardware	Clean & Green Technology
SOAIDEATHON-SH10	**Student Innovation – Clean & Green Technology: Open Challenge**	**Identify and solve a high-impact real-world problem related to waste, water, sanitation, pollution, circularity, biodiversity or climate-responsive local infrastructure that is not already addressed by the four institution-defined problem statements listed under the Clean & Green Technology theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Clean & Green Technology**
SOAIDEATHON-S19	AI-Based Visitor Safety, Crowd and Incident Coordination for Eco and Pilgrimage Sites	Develop a privacy-conscious platform that forecasts crowding, detects unusual incidents and coordinates verified alerts among visitors, site managers and emergency services. It should integrate geofencing, weather and transport conditions, provide multilingual guidance and function in low-connectivity areas without continuously tracking identifiable individuals.	Software	Tourism

SOAIDEATHON-S20	Accessible Journey Planner with Real-Time Barrier and Assistance Mapping	Create a tourism planner for travellers with mobility, visual, hearing, cognitive or age-related needs. It should recommend routes, facilities and time slots based on verified accessibility information, crowds, weather and available assistance. Users and authorized auditors should be able to report temporary barriers with evidence and confidence ratings.	Software	Tourism
SOAIDEATHON-S21	Regenerative Tourism Impact Ledger for Local Communities and Natural Assets	Develop a platform that tracks visitor flows, local purchases, waste, water use, biodiversity pressure and community benefits for a destination. It should help tourists choose lower-impact options and allow local bodies to evaluate whether tourism revenue is supporting conservation and livelihoods. Data claims must include source and verification status.	Software	Tourism
SOAIDEATHON-S22	Verified Generative-AI Destination Copilot with Dynamic Itineraries and Over-Tourism Control	Create a multilingual travel assistant that builds personalized itineraries from verified local information, live weather and transport conditions, accessibility needs, site capacity and community preferences. The system should cite the source of critical facts, clearly mark uncertain or user-generated information, redirect visitors away from overcrowded or sensitive locations and promote locally owned services. It must avoid fabricating opening hours, safety advice or cultural claims.	Software	Tourism
SOAIDEATHON-SH11	**Student Innovation – Tourism: Open Challenge**	**Identify and solve a high-impact real-world problem related to visitor experience, destination safety, accessibility, local livelihoods, hospitality or sustainable tourism that is not already addressed by the four institution-defined problem statements listed under the Tourism theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Tourism**

SOAIDEATHON-H23	AI-Orchestrated Campus Microgrid with Solar, Storage, EV and Critical-Load Resilience	Design a microgrid controller or digital-twin prototype that forecasts renewable generation and demand, schedules batteries and EV charging, protects critical loads during outages and explains dispatch decisions. The system should account for battery health, tariff constraints and uncertainty while allowing manual override by facility operators.	Hardware	Renewable/ sustainable Energy
SOAIDEATHON-S23	Building Energy Flexibility Aggregator for Demand Response and Carbon-Aware Scheduling	Develop a platform that identifies flexible loads such as HVAC, pumps, charging and noncritical laboratory equipment, then recommends schedules that reduce peak demand and carbon intensity without violating comfort or operational constraints. It should quantify user impact, preserve privacy of occupancy data and learn from operator feedback.	Software	Renewable/ sustainable Energy
SOAIDEATHON-H24	Agrivoltaic and Solar-Powered Irrigation Optimization for Small Farms	Design a combined solar generation and irrigation management system that decides when to pump, store energy or defer irrigation based on soil need, weather and electricity availability. The solution should evaluate crop shading, water savings and energy yield, and remain serviceable with locally available components.	Hardware	Renewable/ sustainable Energy
SOAIDEATHON-S24	Community Energy Flexibility and Shared-Battery Digital Twin for Local Renewable Integration	Develop a digital twin that coordinates rooftop solar, shared or second-life batteries, flexible building loads and EV charging across a campus or small community. The platform should forecast generation and demand, test different battery-sharing rules, respect feeder constraints and user preferences, and quantify cost, peak-demand and renewable-utilization impacts. It should prevent any participant from being disadvantaged by opaque allocation rules.	Software	Renewable/ sustainable Energy
SOAIDEATHON-SH12	**Student Innovation – Renewable/ sustainable Energy: Open Challenge**	**Identify and solve a high-impact real-world problem related to renewable generation, storage, energy efficiency, grid integration, clean cooking or community energy that is not already addressed by the four institution-defined problem statements listed under the Renewable/ sustainable Energy theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Renewable/ sustainable Energy**

SOAIDEATHON-S25	Post-Quantum Cryptography Migration Readiness and Crypto-Agility Toolkit	Develop a tool that discovers cryptographic use in institutional applications, certificates, VPNs and embedded devices; assesses exposure to future quantum attacks; and recommends a phased migration to approved post-quantum algorithms. It should support hybrid testing, performance benchmarking, dependency mapping and rollback planning without automatically changing production systems.	Software	Blockchain & Cybersecurity
SOAIDEATHON-S26	Deepfake-Resistant Provenance and Verification System for Official Digital Communications	Create a system that cryptographically signs authorized institutional audio, video, notices and emergency messages and provides a simple verification interface for recipients. It should detect likely manipulation, distinguish unsigned from proven-fake content, preserve privacy and support revocation when a signing credential is compromised.	Software	Blockchain & Cybersecurity
SOAIDEATHON-H25	Zero-Trust Security and Verifiable Device Passport for Campus IoT	Develop a gateway and management platform that continuously authenticates IoT devices, enforces least-privilege communication, detects anomalous behavior and stores a verifiable device passport containing ownership, firmware, vulnerabilities and maintenance history. Compromised devices should be isolated without disabling essential services.	Hardware	Blockchain & Cybersecurity
SOAIDEATHON-S27	Human-Governed Autonomous SOC for Phishing, Identity Abuse and Ransomware Response	Develop an AI-assisted security operations platform that correlates endpoint, identity, email and network events; explains likely attack chains; recommends containment playbooks; and can execute only pre-approved low-risk actions. High-impact actions such as account suspension or network isolation must require human authorization. The system should resist prompt injection and poisoned alerts, maintain evidence provenance and learn from analyst feedback without exposing sensitive logs to external services.	Software	Blockchain & Cybersecurity

SOAIDEATHON-SH13	**Student Innovation – Blockchain & Cybersecurity: Open Challenge**	**Identify and solve a high-impact real-world problem related to cyber defence, privacy, identity, trustworthy digital systems, blockchain or secure infrastructure that is not already addressed by the four institution-defined problem statements listed under the Blockchain & Cybersecurity theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Blockchain & Cybersecurity**
SOAIDEATHON-S28	Curriculum-Aligned Adaptive AI Tutor with Hallucination and Learning-Gap Controls	Develop an AI tutor that uses approved course content, identifies prerequisite gaps, generates adaptive explanations and practice, and cites the source material used. It must detect when it lacks evidence, avoid completing graded work on behalf of students, support multilingual and accessible interaction, and give teachers dashboards focused on misconceptions rather than surveillance.	Software	Smart Education
SOAIDEATHON-S29	Privacy-Preserving Student Success and Well-Being Early-Warning System	Create a system that identifies students who may need academic, financial or well-being support using minimal and consented data. It should explain risk factors, provide supportive intervention options, measure bias across student groups and prevent automated punitive decisions. Students must be able to view and correct key profile information.	Software	Smart Education
SOAIDEATHON-S30	Verifiable Skill Passport and Explainable Internship-Team Matching Platform	Develop a platform that converts verified coursework, projects, competitions and micro-credentials into a portable skill passport and matches students to internships or multidisciplinary teams. Recommendations must explain which evidence supports the match, identify missing skills and avoid ranking based on protected or irrelevant attributes.	Software	Smart Education
SOAIDEATHON-S31	Multimodal Laboratory Skill Assessment and Safety Coach for Outcome-Based Engineering Education	Develop an AI-assisted laboratory platform that observes instrument interactions through approved sensors or video, checks procedural steps against a teacher-defined rubric, detects common safety violations and provides formative feedback after the activity. The system should distinguish evidence from inference, allow instructors to override assessments, avoid facial-recognition-based grading and generate competency evidence aligned with course outcomes rather than a single opaque score.	Software	Smart Education

SOAIDEATHON-SH14	**Student Innovation – Smart Education: Open Challenge**	**Identify and solve a high-impact real-world problem related to learning, assessment, accessibility, educational administration, skill development or student support that is not already addressed by the four institution-defined problem statements listed under the Smart Education theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Smart Education**
SOAIDEATHON-S32	Hyperlocal Multi-Hazard Early-Warning Fusion with Uncertainty-Aware Action Guidance	Develop a platform that combines weather, river, seismic, fire, infrastructure and crowdsourced observations to detect local hazards and issue action-oriented alerts. Each alert should show confidence, affected area, evidence sources and recommended actions for different user groups. The system must avoid alarm fatigue and support authorized correction of false or outdated information.	Software	Disaster Management
SOAIDEATHON-H26	Offline-First Emergency Mesh for Campus and Community Response	Design portable nodes and a mobile application that create a local mesh network for SOS messages, team coordination, resource requests and verified situation reports when cellular or internet services fail. The system should prioritize life-critical messages, prevent spoofing, support multilingual use and synchronize safely when connectivity returns.	Hardware	Disaster Management
SOAIDEATHON-S33	Evacuation Digital Twin for Dynamic Routing of People with Diverse Mobility Needs	Create a building or campus evacuation digital twin that uses occupancy, hazard location and route availability to recommend dynamic evacuation paths. It must account for wheelchair users, temporary injuries, children and crowded bottlenecks, while enabling safety officers to test plans before an incident. The system should remain advisory and fail safely when sensors are unavailable.	Software	Disaster Management

SOAIDEATHON-S34	AI-Guided Post-Disaster Damage Mapping and Relief Prioritization from Drone and Street Imagery	Develop a platform that ingests drone, satellite or street-level images after a flood, cyclone, fire or earthquake to map damaged buildings, blocked routes and likely service disruptions. The system should assign confidence, allow responders to verify or correct findings, combine vulnerability and accessibility information, and recommend inspection or relief priorities without replacing official field assessment. It must record the evidence behind every priority recommendation.	Software	Disaster Management
SOAIDEATHON-SH15	**Student Innovation – Disaster Management: Open Challenge**	**Identify and solve a high-impact real-world problem related to risk reduction, preparedness, early warning, emergency response, recovery or resilient infrastructure that is not already addressed by the four institution-defined problem statements listed under the Disaster Management theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Disaster Management**
SOAIDEATHON-H27	Adaptive Physical-Digital STEM Toy Kit for Neurodiverse and Multilingual Learners	Design a modular toy kit with tangible blocks, sensors and an accessible companion application that adapts STEM challenges to different learning styles and abilities. The kit should provide audio, visual and haptic feedback, work without continuous internet and allow teachers or caregivers to create culturally relevant activities without programming.	Hardware	Games & Toys
SOAIDEATHON-S35	Location-Based Heritage Exploration Game with Verified Cultural Content	Create a mobile or mixed-reality game that guides players through local history, language, art and architecture using missions tied to real places or museum collections. All cultural claims should link to verified sources, and the game should offer safe remote participation for users who cannot travel.	Software	Games & Toys
SOAIDEATHON-H28	Circular Modular Toy Platform from Recovered Materials with Safety Traceability	Design a family of repairable, reconfigurable toys made from safely processed recovered materials. Each component should have a digital material and safety record, standardized connectors and age-appropriate design rules. The project should include a take-back, repair and reconfiguration model rather than a single disposable toy.	Hardware	Games & Toys

SOAIDEATHON-H29	Accessible Haptic-and-Audio Game Controller and Inclusive Play Toolkit	Design a low-cost modular controller and game toolkit for children and adults who cannot comfortably use conventional visual, fine-motor or two-handed controllers. The system should support configurable large buttons, haptics, audio cues, switch access and alternative gestures, and provide an SDK or simple authoring tool so student developers can build inclusive games. The design should emphasize electrical safety, durability and low-cost repair.	Hardware	Games & Toys
SOAIDEATHON-SH16	**Student Innovation – Games & Toys: Open Challenge**	**Identify and solve a high-impact real-world problem related to inclusive play, learning games, toys, game technology, cultural games or safe physical-digital experiences that is not already addressed by the four institution-defined problem statements listed under the Games & Toys theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Games & Toys**
SOAIDEATHON-S36	Evidence-Grounded Civic Grievance Triage and Participatory Budgeting Platform	Develop a multilingual platform that receives civic issues through text, voice or images, identifies duplicates, routes them to responsible units and publicly tracks resolution evidence. Aggregated patterns should help communities propose and prioritize small public works through transparent participatory budgeting, while protecting complainant identity where necessary.	Software	Miscellaneous
SOAIDEATHON-S37	Crowdsourced Accessibility Digital Twin for Public Buildings and Services	Create a platform that maps entrances, lifts, toilets, signage, sensory conditions and service barriers for public buildings. Data should combine structured audits, user reports and computer vision, with confidence and verification status. The platform should recommend prioritized low-cost improvements and provide accessible navigation before and during visits.	Software	Miscellaneous
SOAIDEATHON-S38	Food-Service Demand Forecasting and Surplus Redistribution for Campuses and Hospitality	Develop a system that forecasts meal demand using calendars, weather, attendance and historical consumption; recommends preparation quantities; and safely matches verified surplus with eligible recipients. It should record storage time and temperature, prevent unsafe redistribution and quantify food, cost and carbon savings.	Software	Miscellaneous

SOAIDEATHON-S39	AI-Based Queue, Crowd and Service Experience Optimization for High-Footfall Public Facilities	Develop a privacy-conscious platform for hospitals, campuses, government service centres, event venues or large retail facilities that forecasts demand, estimates waiting time and recommends counter, appointment or space allocation. It should support token-based or anonymous sensing, accessible priority rules, multilingual updates and transparent escalation during unusual crowding. The system must optimize both average waiting time and fairness for vulnerable or time-critical users.	Software	Miscellaneous
SOAIDEATHON-SH17	**Student Innovation – Miscellaneous: Open Challenge**	**Identify and solve a high-impact real-world problem related to a significant societal, institutional, hospitality, retail, entertainment or cross-domain challenge not better covered by another SIH theme that is not already addressed by the four institution-defined problem statements listed under the Miscellaneous theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Miscellaneous**
SOAIDEATHON-S40	Explainable Real-Time Fraud Shield for UPI, Voice Phishing and Social Engineering	Develop a privacy-preserving risk engine that detects suspicious payment behavior, device changes, coercive interaction patterns and voice-phishing indicators before a transaction is completed. It should deliver understandable warnings, support user confirmation without blocking legitimate urgent payments and allow banks or institutions to review false positives.	Software	Fintech
SOAIDEATHON-S41	Consent-Based Alternative Credit and Financial Health Assistant for Students and Micro-Entrepreneurs	Create a platform that helps users understand cash flow, savings capacity and responsible borrowing using only consented data. Any credit-readiness score must be explainable, correctable and separated from protected characteristics. The system should recommend non-credit alternatives, budgeting actions and suitable public schemes instead of encouraging unnecessary debt.	Software	Fintech
SOAIDEATHON-H30	Offline-Capable Inclusive Micro-Payment and Benefit Verification Device	Design a low-cost device and companion software for secure small-value payments or benefit verification in intermittent-connectivity areas. It should support accessible interfaces, transaction limits, tamper evidence, delayed synchronization, duplicate-spend prevention and clear dispute records. The design must not store unnecessary biometric or financial data.	Hardware	Fintech

SOAIDEATHON-S42	Consent-Based MSME Cash-Flow Digital Twin for Invoice Risk and Working-Capital Resilience	Develop a financial-planning platform for micro and small enterprises that creates a cash-flow digital twin from consented invoices, receivables, recurring expenses and payment history. It should forecast liquidity gaps, identify concentration or delayed-payment risk, simulate shocks and compare non-debt, invoice-finance and working-capital options with clear costs. The system must avoid automatic lending decisions, explain every recommendation and allow users to correct imported data.	Software	Fintech
SOAIDEATHON-SH18	**Student Innovation – Fintech: Open Challenge**	**Identify and solve a high-impact real-world problem related to payments, financial inclusion, fraud prevention, compliance, savings, responsible credit or MSME financial resilience that is not already addressed by the four institution-defined problem statements listed under the Fintech theme in this workbook. The team must provide evidence that the problem exists, identify a real stakeholder or user group, compare existing solutions, and build a working software, hardware or integrated prototype with measurable outcomes. Proposals should include safety, privacy, accessibility, sustainability and responsible-AI considerations wherever relevant.**	**Software/Hardware**	**Fintech**